

Teacher Guide — senior / module-15- project-implementation

IKS Curriculum

Table of Contents

Overview

Project Implementation — Capstone

Band: Senior (Grades 9–12) · **Sessions:** 10 × 45 min · **Sanskrit:** Anvayana — Synthesis

Overview

The capstone module of the curriculum. Students choose from a menu of applied projects (or design their own) that draw on concepts from Modules 1–14. They scope, research, design, build, document, and present. Process discipline — daily journal, source citations, honest reflection — is part of the assessment.

NEP 2020 alignment

- §4.6 — Integration of Indian Knowledge Systems
- §4.23 — Experiential / project-based learning
- §11.8 — Holistic, multidisciplinary

Key terms (this module)

- *anvayana* — integration / following-through; capstone synthesis
- *pañcagavya* — five-cow-product preparation (substituted with fermented-foods study for school safety)
- *siddhānta* — concluded principle or doctrine

Prerequisites

Module 2 (Panchabhūta) is recommended before this module. Modules 5 (Bindu Paddhati / dot addition) and 7 (×11 Multiplication) are useful prerequisites for Module 9 and 11.

Materials needed (across 10 days)

- Notebook + pencils per student
- Whiteboard / chart paper
- (See `lesson-plan.md` for per-day material lists)

10-day map

- **Day 1** — Capstone Overview and Project Menu: Review the 10-option project menu; pick a project.
- **Day 2** — Scope and Plan: Write a 1-page project plan: problem, sources, deliverable, deadline.
- **Day 3** — Research — Day 1: Gather sources; cross-reference earlier modules.
- **Day 4** — Research — Day 2: Continue research; sketch initial design.
- **Day 5** — Mid-Project Critique: Pairs swap plans; offer one strength + one concern.
- **Day 6** — Build / Make — Day 1: Start building the artefact (model, poster, deck, garden plot).
- **Day 7** — Build / Make — Day 2: Continue building; daily journal entry.
- **Day 8** — Document: Write up process, sources, reflections (~500 words for Senior).
- **Day 9** — Rehearse Presentation: Pair feedback on final presentation; refine.
- **Day 10** — Project Share Day: Present project to peers + parents; complete self-assessment.

Files in this pack

- `lesson-plan.md` — detailed 10-day arc
- `sources.md` — sources cited in this module
- `teacher-notes.md` — common misconceptions, differentiation
- `parent-guide.md` — what parents should know
- `days/day-NN-*.md` — 10 per-day lesson plans
- `quizzes/formative.md`, `quizzes/summative.md`
- `activities/activity-01-*.md`
- `assessments/rubric.md`, `assessments/project-brief.md`
- `student-workbook.md`
- `writing-prompts.md` (or `oral-recall.md` for Primary)
- `homework.md`
- `slides/deck.md`

Voice

Analytical, comparative, primary-source aware; 20–30 words on average. Pitches the concept first, frames the tradition as historical context, and uses IAST for all Sanskrit terms.

10-Day Lesson Plan

Lesson Plan — Project Implementation — Capstone (Senior)

10-day arc, 45 min per session.

Day 1 — Capstone Overview and Project Menu

Objective: Review the 10-option project menu; pick a project.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 2 — Scope and Plan

Objective: Write a 1-page project plan: problem, sources, deliverable, deadline.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 3 — Research — Day 1

Objective: Gather sources; cross-reference earlier modules.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 4 — Research — Day 2

Objective: Continue research; sketch initial design.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 5 — Mid-Project Critique

Objective: Pairs swap plans; offer one strength + one concern.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 6 — Build / Make — Day 1

Objective: Start building the artefact (model, poster, deck, garden plot).

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 7 — Build / Make — Day 2

Objective: Continue building; daily journal entry.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 8 — Document

Objective: Write up process, sources, reflections (~500 words for Senior).

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 9 — Rehearse Presentation

Objective: Pair feedback on final presentation; refine.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 10 — Project Share Day

Objective: Present project to peers + parents; complete self-assessment.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Pacing notes

- Senior sessions are 45 min. Adjust if your block is different.
- Days 5 and 9 are buffer days — use to reinforce weaker concepts or extend stronger ones.
- The summative project (see [assessments/project-brief.md](#)) runs in parallel from Day 6 onward.

Differentiation

- **For advanced students:** see [teacher-notes.md](#) for extension prompts.
- **For students needing more support:** simplify activities, do more pair-work, allow oral instead of written responses.

Teacher Notes

Teacher Notes — Project Implementation — Capstone (Senior)

Common student misconceptions

- *“This is religious, not scientific.”* — Frame the tradition as historical observation. Pañcabhūta-like classifications of matter exist in many cultures. Indian astronomy and math are documented, peer-reviewed historical fields.
- *“The Sanskrit terms are just jargon.”* — Insist on the triple format (Sanskrit / IAST / meaning) once, then use the Sanskrit alone. The terms are precise; the discipline of using them properly is itself a learning outcome.
- *“If it’s traditional, it must work.”* — Distinguish empirical claims (testable) from ethical/aesthetic frames (which need separate justification). Encourage students to ask “how would I test this?”.

Differentiation strategies

- **Extension for advanced students:** offer one optional “deeper read” per day — usually a primary source or modern paper.
- **Support for students needing scaffolds:** more pair-work, allow oral responses, provide partial outlines for writing prompts.
- **Multilingual classrooms:** Sanskrit terms are universal; vernacular equivalents (Hindi, Tamil, Telugu, Bengali, Marathi etc.) should be welcomed.

Safety

- Any activity involving food, plants, fire, or outdoor work needs explicit safety briefing.
- See the relevant day file for activity-specific safety notes.

Honest framing

- Where modern science clearly extends or revises traditional claims, say so.
- Where the tradition's framing is best understood as ethical/aesthetic, present it as such (not as competing science).
- Where attribution is debated (e.g., the “Vedic” status of Tirthaji's mathematics), present the debate honestly.

End-of-module reflection (for teachers)

- Which concepts landed well? Which didn't?
- Which students grew most? Which need a different approach next module?
- Add notes to `teacher-notes.md` for the next teacher who uses this pack.

Sources

Sources — Project Implementation — Capstone (Senior)

Primary and secondary sources

- All Modules 1–14 of this curriculum
- Project-Based Learning literature (Buck Institute, Edutopia)
- NCERT NEP 2020 §4.23 — experiential learning

A note on citation discipline

Per the curriculum's style guide, citations must be verifiable. If a verse number, page, or attribution cannot be confirmed, the source is paraphrased without invented attribution. Where direct quotes appear in Senior-band day files, the source is identified in IAST + edition.

For deeper reading

- The shared glossary at `curriculum/_shared/glossary.md` contains the IAST + meaning for every Sanskrit term used.
- Other modules in this curriculum that connect: see the module README's “Prerequisites” section.

Day 1 — Capstone Overview and Project Menu

Module: Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: review the 10-option project menu; pick a project.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *anvayana* (IAST: anvayana; lit. “integration / following-through; capstone synthesis”)

Primary source (today’s reading)

From *All Modules 1–14 of this curriculum* — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Capstone overview.** This module is different. No new content delivery. Instead: 10 days of structured project work, applying what you learned from Modules 1–14.
2. **Walk through the 10-option project menu.** Read each, with a 1-minute description per option:
 - Pañcagavya kitchen science → SUBSTITUTED with fermented-foods study (safety)
 - Mandala / Pṛthvī geometry
 - Tithi/festival calendar
 - Herb identification card-deck
 - Magic-square art
 - Yoga sequence design
 - Sustainability audit
 - Doshic self-study
 - Multiplication-method comparison
 - Open-design (your own)
3. **The discipline.** Each project requires: scope, research, design, build, document, present. Daily journal. Source citations.

4. **The safety floor.** Pañcagavya — NEVER cow-urine ingestion in school. Use comparative fermented-foods study (curd, paneer, batter — biochemically interesting and safe).
5. **Today's task.** Read all 10 options in detail (printout provided). DON'T pick yet — sleep on it. Tomorrow you'll choose and scope.

Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Capstone Overview and Project Menu.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 2 — Scope and Plan

Module: Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: write a 1-page project plan: problem, sources, deliverable, deadline.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *pañcagavya* (IAST: pañcagavya; lit. “five-cow-product preparation (substituted with fermented-foods study for school safety)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Each student commits to ONE project.** Show of hands per option to see distribution.
2. **The one-page plan template.**
 - Title (~10 words)
 - Problem (~50 words): what question/need does this address?
 - Sources (~5 listed): the modules + texts you’ll draw from
 - Deliverable (~50 words): what will exist at the end?
 - Timeline (Day 3-10 broken down by phase)
 - Success criteria (3 bullet points: what makes this excellent?)
3. **Work-time.** 30 minutes of silent plan-writing.
4. **Pair-share.** Pairs swap plans, give one strength + one concern (sticky notes).
5. **Submit.** Plans go to the teacher tonight for sign-off. Tomorrow you begin research.

Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Scope and Plan.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 3 — Research — Day 1

Module: Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: gather sources; cross-reference earlier modules.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See *lesson-plan.md* for any day-specific materials)

Vocabulary introduced today

- *siddhānta* (IAST: *siddhānta*; lit. “concluded principle or doctrine”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Research phase begins.** Each student returns to the relevant Modules for content:
 - Doṣa project → Module 4
 - Herbs project → Module 10
 - Astronomy project → Module 6 or 12
 - Math project → Module 5, 7, 9, 11, or 14
 - Ecology project → Module 13
2. **The library / research routine.** Open the relevant pack’s *sources.md*. Open at least 2 cited sources (online or print). Take 2-page notes per source. The notes will support the project’s references.
3. **Critical eye.** Don’t just trust the FIRST source. For each claim, ask: “*Who’s the author? When was this written? What’s their context?*”
4. **Work-time.** 35 minutes of silent research + note-taking.

5. **Sharing (5 min).** Three students briefly describe what they learned and what surprised them.

Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Research — Day 1.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 4 — Research — Day 2

Module: Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: continue research; sketch initial design.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *anvayana* (IAST: *anvayana*; lit. “integration / following-through; capstone synthesis”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Continued research.** Today’s focus: sketches + design.
2. **The design-thinking question.** “*What will this LOOK like when it’s done?*” Draw it. Even a rough sketch helps you find missing pieces.
3. **Source check.** Revisit yesterday’s notes. Are there gaps? Look up one more source if needed.
4. **Work-time.** 30 minutes of sketching + planning the build.
5. **Pair-feedback.** Pairs discuss design. “*What’s the strongest part of your design? What’s still unclear?*”

Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Research — Day 2.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 5 — Mid-Project Critique

Module: Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: pairs swap plans; offer one strength + one concern.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *pañcagavya* (IAST: pañcagavya; lit. “five-cow-product preparation (substituted with fermented-foods study for school safety)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Critique circle structure.** Pairs swap project plans + sketches. 5 minutes silent reading. Then each gives:
 - One strength (sticky note).
 - One concern (sticky note).
 - One question (verbal).
2. **Pairs rotate twice.** Each student gets feedback from 3 different partners.
3. **Synthesise.** Each student writes a 100-word “what I’m changing” note based on critique. May include:
 - Narrowing scope
 - Adding a source
 - Clarifying deliverable
 - Revising timeline
4. **Honest about limits.** “*What’s still hard? What might fail? What’s your Plan B?*” Real projects always have unknowns. Acknowledge them.
5. **Submit revised plan.** Teacher reviews tonight. Day 6 = build phase begins.

Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Mid-Project Critique.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 6 — Build / Make — Day 1

Module: Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: start building the artefact (model, poster, deck, garden plot).

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *siddhānta* (IAST: *siddhānta*; lit. “concluded principle or doctrine”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Build phase begins.** Materials should be ready (announced last week). If not, today is also a “gather materials” day.

2. Project-specific guidance per option.

- Magic-square art → grid + colours + calligraphy.
- Herb deck → photographs / drawings + card layout + lamination.
- Yoga sequence → flow design + safety check + pose drawings.
- Sustainability audit → measurement instruments + spreadsheet.
- And so on.

3. **Daily journal.** Today's entry: "*What did I build today? What worked? What didn't?*"

4. **Work-time.** 35 minutes of focused making.

5. **End-of-day share (5 min).** Three students share progress.

Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Build / Make — Day 1.

Wrap-up (5 min)

- One-sentence exit ticket: "*What's one thing you learned today?*"
- Preview tomorrow.

Homework

(See [homework.md](#) for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 7 — Build / Make — Day 2

Module: Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: continue building; daily journal entry.

Materials

- Whiteboard / chart paper

- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *anvayana* (IAST: anvayana; lit. “integration / following-through; capstone synthesis”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Continue building.** Focus on completing the major deliverable.
2. **Journal entry today:** “*What progress? What’s slowed me down? What do I still need to do tomorrow?*”
3. **Mid-build review.** Teacher checks each student’s progress. Flags students who may need help in time-management or scope.
4. **Honest scope adjustment.** If you realise you can’t finish in time, narrow the scope NOW (not on Day 9). Better to do less, well, than more, poorly.
5. **Independent work-time.** 40 minutes. Quiet, focused.

Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Build / Make — Day 2.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.

- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 8 — Document

Module: Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: write up process, sources, reflections (~500 words for senior).

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *pañcagavya* (IAST: *pañcagavya*; lit. “five-cow-product preparation (substituted with fermented-foods study for school safety)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Documentation phase.** Today: write up your project.
2. **The structure (Senior — ~500 words).**
 - **Problem & motivation** (~100 words): what did you set out to do?
 - **Sources used** (~100 words): which Modules, which texts, which observations?
 - **Process** (~150 words): what did you actually do? Day by day?
 - **Findings / outcomes** (~100 words): what did you produce / observe?
 - **Reflection** (~50 words): what surprised you? What would you do differently?
3. **Honest reflection rule.** Include at least ONE thing that didn’t work. Real research has failures. Including yours is intellectually honest — and pedagogically valuable.
4. **Citations.** Use the format: *Author, Title, Year*. See pilot modules’ `sources.md` for examples.
5. **Drafting time.** 35 minutes silent writing. Teacher available for questions.

Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Document.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 9 — Rehearse Presentation

Module: Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: pair feedback on final presentation; refine.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *siddhānta* (IAST: *siddhānta*; lit. “concluded principle or doctrine”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Pair-rehearsal.** Pairs swap presentations. Time the 5-minute talk. Friend gives:
 - One strength (clear, well-organised, engaging).
 - One specific suggestion (slow down, add an example, more eye contact).
2. **Refine.** Each student adjusts their presentation based on feedback. 10 minutes solo refinement.
3. **Group-rehearsal (5 students share with class).** Selected by teacher (varied projects). Class gives 2 minutes of feedback each.
4. **Logistics check.** Materials ready for tomorrow? Backup plan if something breaks?
5. **Self-assessment.** Each student rates themselves on the rubric (assessments/rubric.md). They submit this PRE-presentation; the teacher will compare with their own scoring after the share.

Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Rehearse Presentation.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 10 — Project Share Day

Module: Project Implementation — Capstone · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: present project to peers + parents; complete self-assessment.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *anvayana* (IAST: anvayana; lit. “integration / following-through; capstone synthesis”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Project share day.** Each student presents their project (5 minutes including Q&A). Audience: peers, optionally parents, optionally other teachers.
2. **Format.** Brief: what + why + how + what-I-learned. Show the artefact. Take 1-2 questions.
3. **Audience role.** Each peer takes one note per project (1 sticky note, one observation or compliment). At the end, sticky notes go to each presenter.
4. **Self + teacher assessment.** After the share, each student completes their final self-assessment on the rubric. The teacher fills the corresponding teacher-assessment. Within a week, the two are compared in a 1-1 conversation.
5. **Module close (5 min).** Whole-class reflection: “*Looking back across all 15 modules — Modules 1-14 of curriculum, Module 15 of project — what’s ONE thing that will stay with you?*” Each student says one sentence.
6. **Curriculum close.** Brief teacher comment + final round of applause. The IKS curriculum is complete.

Activity (15 min)

A scaled version of *Mid-Project Critique Circle* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Project Share Day.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Formative Quiz — Project Implementation — Capstone (Senior)

When: mid-module (after Day 5). **Duration:** 25 minutes. **Format:** mix of short answer, multiple choice, and reflection.

Questions

1. Name the three / four key terms introduced in this module so far. Give the IAST and one-line meaning of each.
2. State the central idea of Project Implementation — Capstone in one sentence.
3. (Multiple choice) Which day in this module focused on ‘Research — Day 1’? (a) Day 1 (b) Day 2 (c) Day 3 (d) Day 4
4. True / False — and explain in one sentence: *Project Implementation — Capstone is purely a religious concept and has no scientific framing.*
5. Connect Project Implementation — Capstone to **one** other module in this curriculum. Which module, and how?
6. List **two** sources cited so far in this module.
7. (Short answer) Why does the module distinguish ‘anvayana’ from related concepts? Give one example.
8. (Short answer) In your own words, explain Day 3’s concept.
9. Identify **one** common misconception about Project Implementation — Capstone and rebut it in one sentence.
10. Write **one** question you have about this module so far — something you want clarified before the summative.

How this is used

This quiz is **formative** — it's a check-in to see what's landing, not a grade. Teacher reviews answers to adjust pacing for Days 6–10. Students may swap and self-mark.

Summative Quiz — Project Implementation — Capstone (Senior)

When: end of module (after Day 10). **Duration:** 45 minutes. **Total points:** 30.

Questions

Part A — Vocabulary (~5 points) A1. Define *anvayana* in your own words. A2. Define *pañcagavya* in your own words. A3. Define *siddhānta* in your own words.

Part B — Concepts (~10 points) B1. Explain the central idea of Project Implementation — Capstone in your own words (3–4 sentences). B2. Describe the method or framework students used in Day 4 of this module. B3. Identify ONE source cited in this module and explain why it matters. B4. Compare Project Implementation — Capstone with a concept from another module (your choice). What's similar, what's different? B5. Identify ONE common misconception about this module and rebut it in 2–3 sentences.

Part C — Application (~5 points) C1. Give an example of Project Implementation — Capstone from your own daily life. C2. Describe one way this module connects to a current real-world issue (or scientific question).

Part D — Reflection (~5 points; ungraded for content) D1. What surprised you most in this module? D2. What's one question you still have?

Part E — Primary source (~5 points; Senior only) E1. Quote or paraphrase one passage from a primary source cited in this module. Explain it in your own words.

Marking guidance

- Vocabulary: give credit for correct meaning, even if exact wording differs.
- Concepts: look for evidence of understanding, not memorisation.
- Application: any reasonable example accepted.
- Reflection: ungraded for content; awarded for thoughtfulness.

Pass mark

A score of 18/30 indicates the student has met the module's learning objectives.

Homework

Homework — Project Implementation — Capstone (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — Capstone Overview and Project Menu** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Scope and Plan** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Research — Day 1** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — Research — Day 2** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Mid-Project Critique** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Build / Make — Day 1** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Build / Make — Day 2** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Document** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Rehearse Presentation** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Project Share Day** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.

Writing Prompts

Writing Prompts — Project Implementation — Capstone (Senior)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 300-word response.

1. In 3–4 sentences, explain Project Implementation — Capstone to a friend who hasn't taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Project Implementation — Capstone with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?

5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Project Implementation — Capstone? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?
9. Read one primary-source passage cited in this module. Translate or paraphrase it; comment on what it adds to your understanding.
10. Identify a contemporary debate about the IKS framework presented here (e.g., is Tirthaji's mathematics genuinely Vedic? are doṣas a valid framework today?). Take a position with two pieces of evidence.

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Activity 1 — Mid-Project Critique Circle

Module: Project Implementation — Capstone · **Band:** Senior · **Duration:** 30 min

Purpose

Reinforce the day-by-day concepts of Project Implementation — Capstone through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

student project plans, sticky notes

What students do

Pairs swap project plans, read in 5 minutes, then offer one strength (one sticky note) and one concern (another sticky note). Pairs rotate twice. Refine plan based on feedback.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (20 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Project Brief

Project Brief — Capstone Project (Student Choice)

Module: Project Implementation — Capstone · **Band:** Senior

What you will do

Choose one of the 10-option project menu (see README.md) or pitch your own. Deliverables: 1-page plan, daily journal, final artefact, ~500-word reflection (Senior; ~300 for Middle; oral share for Primary), 5-minute presentation. Safety: NO cow-urine ingestion for any pañcagavya project — pivot to fermented-foods comparative study instead.

Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a poster / model / digital deliverable).
- **Written reflection** (500 words (Senior)): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (5-minute share with the class).

Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.

- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

Sources

You must cite at least 3 sources. Use the form: *Author, Title, year*. (See `../sources.md` for examples.)

Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.
- **Capstone safety:** if you choose a pañcagavya project, you **MUST** use the fermented-foods comparative-study substitution — cow-urine ingestion is **NOT** allowed.

Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.

Rubric

Assessment Rubric — Project Implementation — Capstone (Senior)

This rubric applies to the project (see `project-brief.md`) and may also inform the summative quiz.

Criteria

Accuracy of concepts — uses Sanskrit terms correctly, gets the central idea right

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Source discipline — cites sources, distinguishes traditional framing from empirical claim

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Application — connects the module to other modules and to real-world examples

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Presentation — clear, organised, evidence-supported

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Reflection — honest, thoughtful, identifies what was learned and what remains uncertain

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

How the teacher uses this rubric

1. After the project share, score each student on each criterion.
2. Convert to a band (4 = Exceeding, 3 = Meeting, etc.) for the report card.
3. Share at least one strength and one area for growth with each student.

Self-assessment (Senior students)

Senior students complete a parallel self-assessment using the same rubric before the teacher's assessment. The teacher and student then meet to compare.